

Assignment: Hibernate ORM Implementation

Objective: Develop a Hibernate-based Java application to manage an e-commerce system with the following entities: **Category, Product, Users, Orders, and OrderDetails**. Implement relationships among these entities and configure Hibernate to persist data in a relational database.

Tasks:

1. Set Up Hibernate Project:

- Create a Maven/Gradle project.
- Add dependencies for Hibernate, MySQL (or H2 for in-memory database), and any required libraries.
- Configure `hibernate.cfg.xml` or `application.properties` for database connection.

2. Define Entities & Relationships:

1. Category Entity:

- `id` (Primary Key, auto-generated)
- `name` (Unique, Not Null)
- `description`
- Relationship: One-to-Many with **Product**

2. Product Entity:

- `id` (Primary Key, auto-generated)
- `name` (Not Null)
- `price` (Decimal, Not Null)
- `stockQuantity` (Integer)
- Relationship: Many-to-One with **Category**

3. Users Entity:

- `id` (Primary Key, auto-generated)
- `username` (Unique, Not Null)
- `password` (Hashed, Not Null)
- `email` (Unique, Not Null)
- `role` (ADMIN, CUSTOMER)
- Relationship: One-to-Many with **Orders**

4. Orders Entity:

- `id` (Primary Key, auto-generated)
- `orderDate` (Timestamp, Not Null)
- `totalAmount` (Decimal, Not Null)
- Relationship: Many-to-One with **Users**, One-to-Many with **OrderDetails**

5. OrderDetails Entity:

- `id` (Primary Key, auto-generated)
- `quantity` (Integer, Not Null)
- `unitPrice` (Decimal, Not Null)
- Relationship: Many-to-One with **Orders**, Many-to-One with **Product**

3. Implement Entity Mappings:

- Use **JPA Annotations** (`@Entity`, `@Table`, `@Id`, `@GeneratedValue`, `@ManyToOne`, `@OneToMany`, etc.).
- Ensure **cascading** and **fetching strategies** are properly defined.

4. Configure Hibernate and Test Database Operations:

- Implement **HibernateUtil.java** to manage `SessionFactory`.
- Perform CRUD operations:
 - Insert new Categories, Products, Users.
 - Create Orders with multiple OrderDetails.
 - Fetch Orders along with associated Users and Products.

5. Bonus (Optional Enhancements):

- Implement **Named Queries** for fetching products by category.
- Implement **criteria queries** using `CriteriaBuilder`.
- Add **soft delete** functionality using a `deleted` flag instead of physical deletion.
- Implement **pagination** for product listings.

Submission Guidelines:

- Submit a **GitHub repository link** containing:
 - Java source files (`.java`)
 - `pom.xml` or `build.gradle`
 - Database schema or `schema.sql`
 - Test cases for CRUD operations
- Include a **README.md** with setup instructions and execution guide.

Evaluation Criteria:

- Correctness and completeness of entity relationships
 - Proper use of Hibernate ORM annotations and mappings
 - Functional CRUD operations with correct data persistence
 - Code quality, readability, and best practices
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